

# Archival ZTF streak photometry of retired GEO satellites

**Yusuf Enes Korkmaz**

Independent researcher, Hamburg, Germany

Email: yusufeneskorkmaz@outlook.com · ORCID: 0009-0008-0169-8611

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## Abstract

Public wide-field survey images can provide incidental photometry of retired geosynchronous (GEO) satellites without dedicated observing time. We audited Zwicky Transient Facility (ZTF) exposure metadata, then inspected public science and mask cutouts for two retired GOES spacecraft. A fixed, stratified image sample contained a streak at the predicted location in all 80 selected exposures (40 GOES 8 and 40 GOES 12). The median integrated aperture signal-to-noise ratio (SNR) was 185 and 183, respectively. Field-star zero points were available for every exposure, but these detections do not establish a survey-wide completeness function or an attitude solution. A preregistered GOES 12 search did not recover published radar periods. The event-based change experiment produced one evaluable negative result and two event cases that could not support the fixed decision rule. The evidence favors within-exposure analysis for sufficiently fast, repeatedly modulated rotators; slowly varying or evolving tumblers require denser temporal coverage.

## 1. Motivation and scope

Retired GEO spacecraft can change rotation state over years, while conventional light-curve campaigns sample a small fraction of that history. Wide-field archives offer a different sampling pattern: many serendipitous 30 s visits distributed over long baselines. For a GEO object, the apparent motion in a sidereally tracked ZTF exposure produces a trail of roughly 440 pixels. Position along a resolved trail can therefore encode sub-second *relative* brightness samples even though the shutter was open for 30 s. This does not make the archive a high-cadence monitoring program: most nights are sparse, and linking rotational phase across nights remains difficult.

The four underlying reports are unpublished internal reports available from the author on request. They answer different questions. The metadata audit counts predicted footprint intersections, not detections. The first image check measures whether bright GOES trails are actually present and extractable. The locked GOES 12 period search tests recovery against independently published radar periods. The locked change experiment tests a per-streak variability proxy around public anomaly dates. None of these analyses estimates a calibrated spin-axis posterior or infers that an operator-reported anomaly changed pointing. The values below retain their original denominators and decision rules. [1–4]

## 2. Data and method

### Archive opportunity and image selection

The metadata audit matched predicted GEO positions at ZTF exposure midpoints to public science-quadrant footprints, with Sun illumination, horizon, and full-trail containment checks. Across six retired GEO targets it found **6,695 predicted object–exposure intersections**. GOES 12 accounted for **1,336** midpoint intersections on **488** Palomar evenings. These counts describe geometric opportunities only. The image check then selected 40 exposures each for GOES 8 and GOES 12, spread across 20 nights and the years 2019–2026. Selection was conditioned on predicted coverage and data-quality metadata, so the observed 80/80 detection fraction must not be generalized to all survey exposures. [1,2]

Science images and associated masks were obtained as public cutouts through the NASA/IPAC Infrared Science Archive (IRSA). Historical orbital predictions were propagated with SGP4 to place the expected line and select cutouts; the note does not reproduce orbital elements. Measured streaks were registered close to the predicted direction and position. The predicted lengths in the selected image sample were 425–448 pixels. The cutout analysis enlarged marginal windows to avoid truncating the full trail. [2]

### Streak photometry and quality control

A seeing-scaled cross-track aperture followed each registered line. Adjacent sidebands supplied a local background estimate; three-pixel bins along the trail yielded a median relative sample spacing of **0.205 s** for both objects. The observed endpoints were mapped onto the 30 s shutter interval. This supports sub-second *relative* timing, while a partially illuminated streak can leave the absolute origin uncertain by several seconds. SNR here is an exploratory integrated aperture estimate from sideband noise, FITS gain, and source shot noise, not the survey pipeline's point-source SNR. [2]

Bad or saturated pixels and bright-star crossings were flagged. The mask bit denoting an aircraft or satellite track was retained as a trace label rather than treated as a bad pixel. In the first conservative screen, **19 of 40** streaks per object were wholly clean for short-period analysis. A later preregistered search masked only affected *segments*, requiring at least 90 usable bins, 18 s of span, separated groups, and integrated SNR at least 10; this recovered eight otherwise rejected streaks among the 16 eligible GOES 12 epoch-window streaks. These are distinct cohorts and should not be added together. [2,3]

Every one of the 80 science-image headers had usable ZTF field-star zero-point information. Background-subtracted counts were converted with  $m = \text{MAGZP} - 2.5 \log_{10}(\text{DN})$  after rescaling a bin to 30 s equivalent flux. The large-aperture correction was within 0.05 mag of zero for all 80. No satellite color term was applied, so absolute values are color-uncorrected ZTF-band estimates. A free offset per streak was used in the cross-night period fit; its rankings therefore did not depend on the image zero point. The header's zero-point RMS describes field-star calibration scatter, not a full satellite photometric uncertainty budget. [2,3]

3. Measured streaks and light curves

| Object  | Selected exposures / nights | Detected | Wholly clean | Integrated SNR, median (range) | Median matched stars | Median zero-point RMS |
|---------|-----------------------------|----------|--------------|--------------------------------|----------------------|-----------------------|
| GOES 8  | 40 / 20                     | 40       | 19           | 185 (33–2121)                  | 489                  | 0.030 mag             |
| GOES 12 | 40 / 20                     | 40       | 19           | 183 (23–2424)                  | 840                  | 0.032 mag             |

Table 1. Results for the stratified, predicted-streak image sample. The calibration columns summarize ZTF header diagnostics; SNR is the research aperture estimate. Median integrated, color-uncorrected magnitudes were 12.90 for GOES 8 and 13.16 for GOES 12. [2]

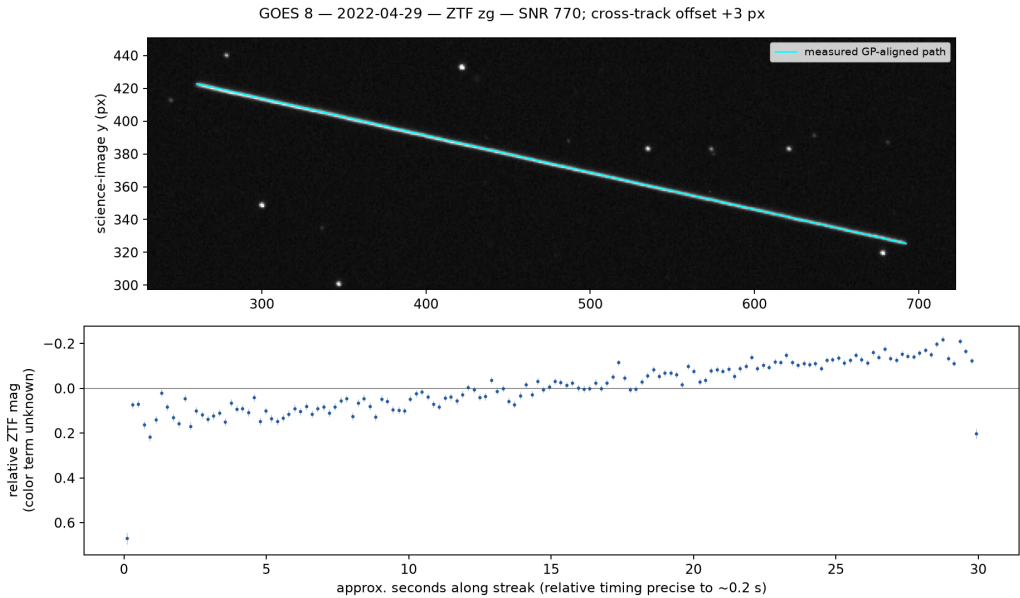


Figure 1. GOES 8 on 2022-04-29. The upper panel shows the measured line on the science cutout. The lower panel uses relative magnitude; the isolated endpoint samples and the gradual trend should not be mistaken for a repeated spin cycle. Image and photometry from the exploratory real-image report. [2]

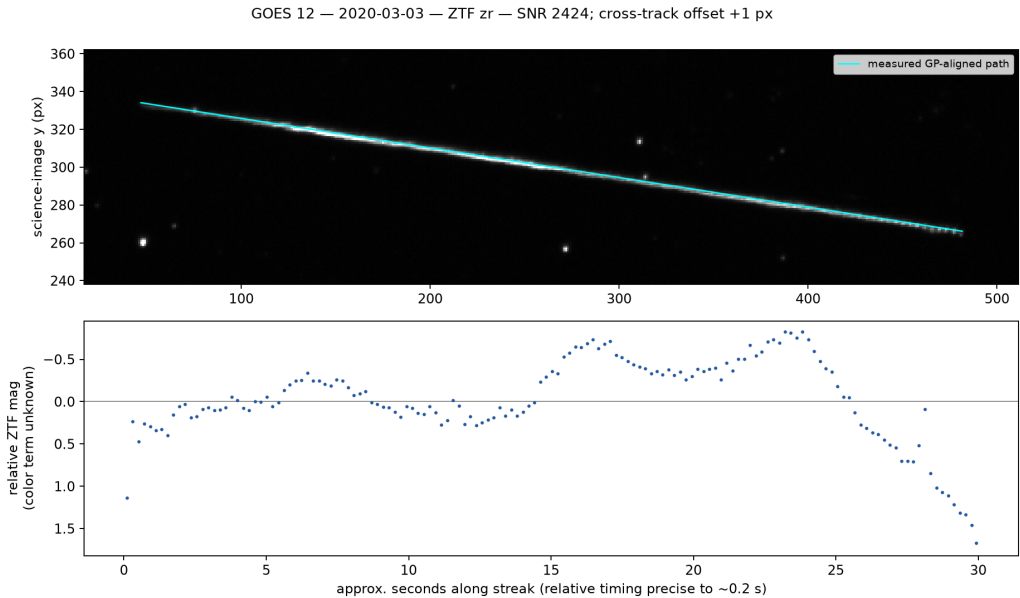


Figure 2. GOES 12 on 2020-03-03. Brightness structure is clear within one exposure, but the approximately 30 s span is short compared with the published minute-scale periods. The relative time coordinate is more secure than its absolute shutter-time origin. Image and photometry from the exploratory real-image report. [2]

4. Negative tests

GOES 12 period recovery

The period-search method, image eligibility, 5–7200 s grid, and ranked candidates were locked before inspecting the published comparison periods. Of 22 unique predicted science-image candidates in the combined windows, three public products were unavailable; 19 science/mask pairs were analyzed and 16 streaks met the segment-aware rule. The December 2019 epoch had only four eligible streaks after the permitted window expansion, below the fixed eight-streak minimum, and received no ranking. The overlapping February 2020 windows had 12 and 9 eligible streaks; they are not independent replications. [3]

| Radar epoch | Published GOES 12 period | Top locked candidate | Registered result                            |
|-------------|--------------------------|----------------------|----------------------------------------------|
| 2019-12-06  | 882 s                    | no ranking           | unevaluable: 4 eligible streaks              |
| 2020-02-18  | 462 s                    | 40.703 s             | no direct or prespecified simple-ratio match |
| 2020-02-20  | 454 s                    | 42.482 s             | no direct or prespecified simple-ratio match |

Table 2. Published values are radar synodic long-axis precession/rotation periods; the locked candidates come from a common, constant-period optical phase-curve fit. They should not be relabeled as equivalent period kinds. No top-three candidate matched either February radar value or its prespecified 1/3, 1/2, 2, or 3 ratio within 5%. [3,7]

The minimum reduced chi-squared was **429** and **519** in the February windows, indicating that the fixed coherent phase curve plus propagated pixel-noise model did not describe the measurements. Published GOES 12 radar periods moved from 882 s in December to 462–454 s in February; even the two February values differ over two days. This rapid evolution, together with four-to-six-day offsets from the radar epochs and sparse survey nights, prevents a credible constant-period phase link across the optical visits. Model inadequacy and timing uncertainty also matter, so evolution is a strong limitation rather than a uniquely proven cause of every short-period fit minimum. The preregistered search **did not recover** the radar periods. [3,7]

The separate within-streak 0.5–15 s scan of the 38 wholly clean exposures produced five nominal interior peaks with false-alarm probability below 0.01, but no peak repeated on another clean night for the same object. Isolated glints and steps violate the white-residual assumption behind those nominal probabilities. **Confirmed sub-15 s periods: zero for both objects.** [2]

Event-based change detection

The preregistered proxy classified a streak as variable only when its robust, detrended magnitude spread divided by median bin uncertainty exceeded a threshold set from 181 clean control streaks. Controls also supplied their own stability check, so that check is in-sample. The intended event-based change claim was **not evaluable for two of three event objects**: Intelsat 29e had no selected footprint intersection in its fixed window; Eutelsat 113 West A had seven clean post-event streaks, below the required eight. Galaxy 15 was evaluable and **negative**: 0/64 variable before versus 2/75 after, Fisher two-sided  $p = 0.4995$ , with no accepted change point. No event satellite yielded a positive date-localized change. This is a variability proxy, not a direct attitude-state test. [4]

## 5. What the archive supports

Resolved streaks can act as short time series when the object is bright, the trail is well contained, and a rotational signature repeats several times within a single exposure. This is the regime where sub-second along-trail sampling has the strongest interpretive value. Even there, adjacent bins share point-spread-function information, bright-star crossings need local masking, and isolated glints should not be reported as periods without repeatability or an adequate noise model. [2,3]

Slow tumblers and objects with evolving spin need denser, more contemporaneous sampling than these sparse ZTF nights provided. A 30 s streak covers only a small fraction of a roughly 450–900 s cycle, and cross-night phase coherence requires the period and its drift to be constrained. A synthetic GOES 12 identifiability exercise produced optimistic *local*, *correct-mode* axis-angle bounds, but did not solve the global period–drift–axis ambiguity; it is not a measured GOES attitude result. Future work should treat period evolution, phase-cycle ambiguity, absolute timing, and correlated photometric error explicitly before attempting state inference. [1,3]

## 6. Related work

Karpov and Peloton (2022) detected repeated satellite glints in the ZTF alert stream, including objects on GEO-like orbits. Their alert-tracklet method demonstrates the value of a single exposure for *fast* flashing objects; this note instead measures continuous flux along predicted science-image trails and emphasizes non-recovery when cycles are not repeated. [5] Rachith, Hellmich, Irureta-Goyena, and Kneib (2023) argued for extracting debris tracks and photometry from astronomical archives, with VST/OmegaCAM as a principal case. Their archive-reuse motivation is closely related, but it does not validate this ZTF sample or its period estimates. [6]

Benson and colleagues combined radar and optical measurements of retired GOES spacecraft. Their 2020 report supplies the dated GOES 12 radar periods used only *after* the candidate list was locked; it also describes GOES 8 as tumbling and GOES 12 as closer to uniform rotation in the February observations. Their 2021 GOES 8 optical folding and radar precession measurements illustrate why period kind, epoch, and method must be stated before comparison. [7,8]

## 7. Data availability and acknowledgment

A light-curve archive and an accompanying extraction tool are in preparation. The present note summarizes results in the four research reports; it does not release image cutouts, orbital elements, per-bin photometry, or a certified attitude product. The public ZTF images are available from IRSA's ZTF Image Service (DOI: 10.26131/IRSA539). [9]

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## References and source reports

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